Week 6, Lecture 2
The Tavis–Cummings Model and Superradiance

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Office hours: **This Week Thursday 1-2 pm**

Recap: The Jaynes–Cummings model

One two-level atom interacts with one quantised cavity mode:

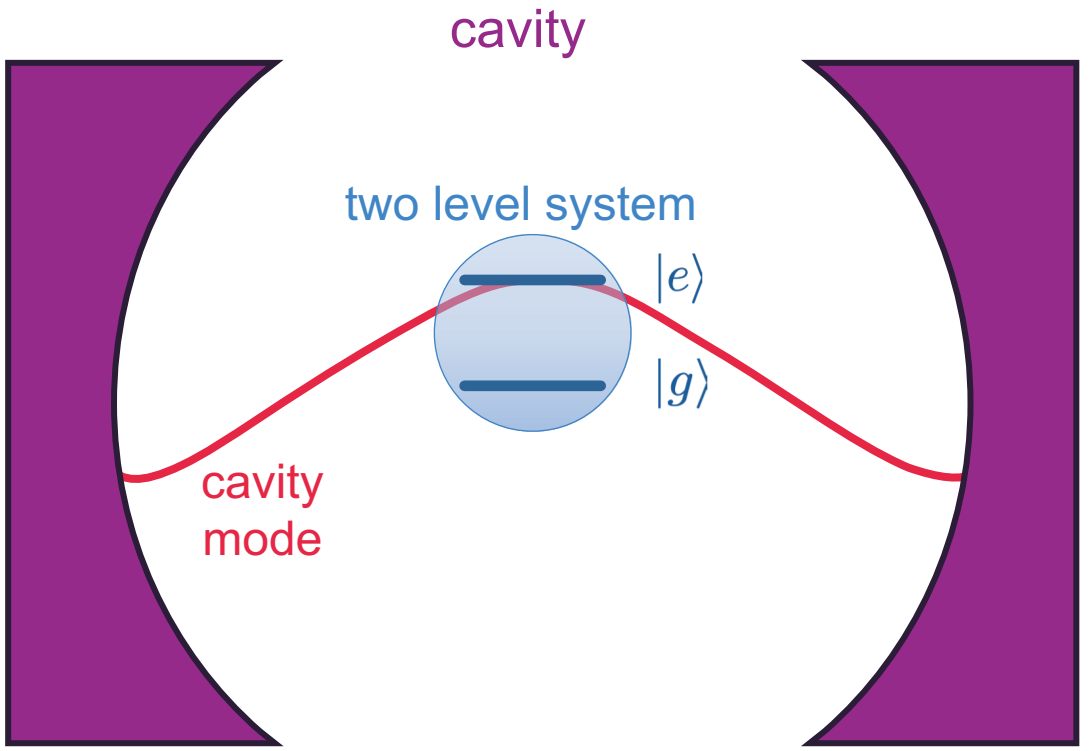
$$H_{\mathrm{JC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a \sigma_+\sigma_- + \hbar g\left(a\sigma_+ + a^\dagger\sigma_-\right)$$

The interaction exchanges an excitation between the atom and cavity:

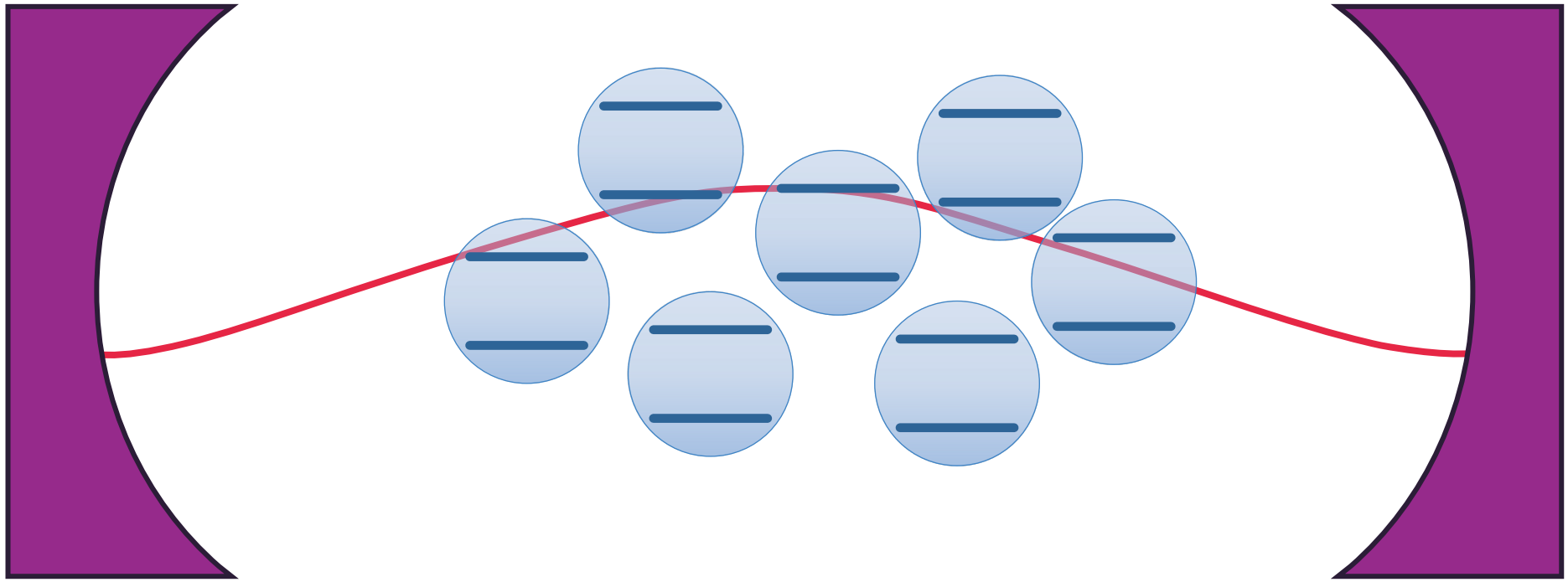
$$|e,n\rangle \longleftrightarrow |g,n+1\rangle$$

The total number of excitations is conserved:

$$\hat{N}_{\mathrm{exc}} = a^\dagger a + \sigma_+\sigma_-$$



In this lecture



What happens when N atoms all couple to the same cavity mode?

What you will learn today

How the Jaynes–Cummings model extends to many emitters

Why symmetric and antisymmetric atomic states couple differently to a common field

How collective atomic emission produces a superradiant burst

From One Atom to Many

Collective light–matter interactions

- The Jaynes–Cummings model describes one two-level atom interacting with one quantised cavity mode.
- We now consider many atoms coupled to the same electromagnetic field.
- A shared field can make the atoms behave collectively rather than independently.
- Collective behaviour can enhance emission, suppress emission, or create states that do not couple to the field.

$$\text{Jaynes–Cummings} \longrightarrow \text{Tavis–Cummings} \longrightarrow \text{superradiance and subradiance}$$

Many Atoms Coupled to One Cavity Mode

Consider N identical two-level atoms inside the same cavity:

- Each atom has transition frequency ω_a.
- All atoms interact with the same cavity mode of frequency ω_c.
- $\sigma_{\pm}^{(i)}$ act only on atom i.
- We initially assume that every atom has the same coupling strength g.

$$H = \hbar\omega_c a^\dagger a + \hbar\omega_a \sum_{i=1}^{N} \sigma_+^{(i)}\sigma_-^{(i)} + \hbar g \sum_{i=1}^{N} \left(a\sigma_+^{(i)} + a^\dagger \sigma_-^{(i)} \right)$$

Every atom interacts with the **same operator** a and therefore with the same photon mode.

Collective Atomic Operators

Instead of describing each atom separately, define operators for the entire ensemble:

$$J_+ = \sum_{i=1}^{N} \sigma_+^{(i)}, \qquad J_- = \sum_{i=1}^{N} \sigma_-^{(i)}$$

$$J_z = \frac{1}{2} \sum_{i=1}^{N} \sigma_z^{(i)}$$

Ignoring an additive constant, the Hamiltonian becomes

$$H_{\mathrm{TC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a J_z + \hbar g \left(a J_+ + a^\dagger J_- \right)$$

This is the **Tavis–Cummings Hamiltonian**.

When Does Collective Coupling Occur?

For atoms with individual couplings g_i,

$$\hat{H}_I = \hbar \sum_{i=1}^{N} \left(g_i \hat{a} \hat{\sigma}_+^{(i)} + g_i^* \hat{a}^\dagger \hat{\sigma}_-^{(i)} \right)$$

The individual-emitter Hamiltonian becomes collective when the emitters are identical and experience the same complex coupling:

$$\omega_i = \omega_a, \qquad g_i = g$$

We can then define

$$\hat{J}_\pm = \sum_{i=1}^{N} \hat{\sigma}_\pm^{(i)}, \qquad \hat{J}_z = \frac{1}{2} \sum_{i=1}^{N} \hat{\sigma}_z^{(i)}$$

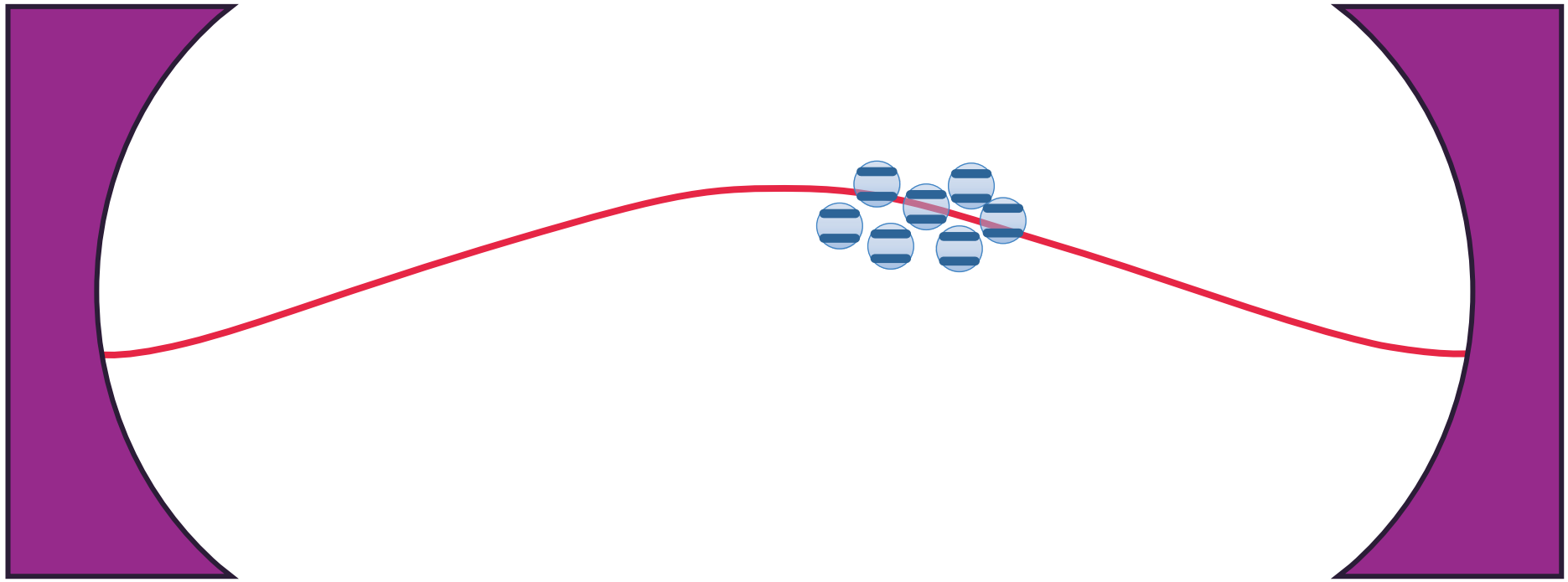
giving the Tavis–Cummings Hamiltonian

$$H_{\mathrm{TC}} = \hbar \omega_c a^\dagger a + \hbar \omega_a J_z + \hbar g \left(a J_+ + a^\dagger J_- \right)$$

When is $g_i \simeq g$?

- The emitters couple to the same field mode.
- They experience approximately the same field amplitude and phase.
- Their transition dipoles have the same strength and orientation.
- Their transition frequencies are approximately equal.

One way to obtain collective coupling



The field interacts with every atom in approximately the same way.

Total Excitation Number Is Conserved

Define the total excitation-number operator:

$$\hat{N}_{\mathrm{exc}} = a^\dagger a + \sum_{i=1}^{N} \sigma_+^{(i)} \sigma_-^{(i)} = a^\dagger a + J_z + \frac{N}{2} \qquad \sigma_+^{(i)} \sigma_-^{(i)} = |e_i\rangle\langle e_i|$$

For the Tavis–Cummings Hamiltonian,

$$\left[H_{\mathrm{TC}}, \hat{N}_{\mathrm{exc}} \right] = 0$$

Therefore, the interaction can move an excitation between the atoms and cavity, but it cannot change the total number:

$$\text{one photon} \longleftrightarrow \text{one atomic excitation}$$

States with different total excitation numbers belong to different manifolds and do not couple to one another.

Two Atoms Sharing One Excitation

For two atoms, the one-excitation manifold contains three basis states:

$$|eg,0\rangle, \qquad |ge,0\rangle, \qquad |gg,1\rangle$$

Here, the first two labels describe the states of atoms 1 and 2, while the final label gives the number of cavity photons.

The interaction Hamiltonian is

$$H_I = \hbar g \left(a J_+ + a^\dagger J_-\right)$$

coupling doesn't involve individual spins, only collective

Its action on the two atomic-excitation states is

$$H_I|eg,0\rangle = \hbar g|gg,1\rangle$$

$$H_I|ge,0\rangle = \hbar g|gg,1\rangle$$

Symmetric and Antisymmetric States

Construct symmetric and antisymmetric combinations of the two possible atomic excitations:

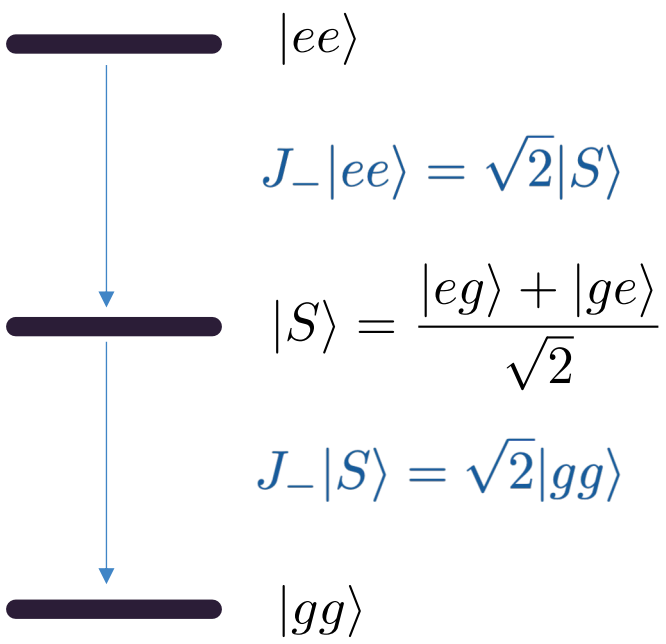
$$|S\rangle = \frac{|eg\rangle + |ge\rangle}{\sqrt{2}} \qquad |A\rangle = \frac{|eg\rangle - |ge\rangle}{\sqrt{2}}$$

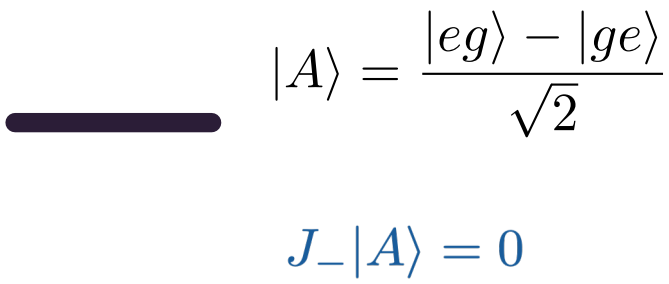
Their coupling to the cavity is

$$H_I|S,0\rangle = \sqrt{2}\hbar g\,|gg,1\rangle \qquad H_I|A,0\rangle = 0$$

- $|S\rangle$ is the symmetric **bright state**.
- $|A\rangle$ is the antisymmetric **dark state**.

Two-Atom Dicke Ladder





Two Atoms Form an Enhanced Emitter

Within the bright subspace

$$\{|S,0\rangle,\ |gg,1\rangle\}$$

$$|S\rangle = \frac{|eg\rangle + |ge\rangle}{\sqrt{2}}$$

$$H_{\mathrm{TC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a J_z + \hbar g\left(aJ_+ + a^\dagger J_-\right)$$

$$H_{\mathrm{TC}}|S,0\rangle = \hbar\omega_a|S,0\rangle + \sqrt{2}\hbar g\,|gg,1\rangle$$

$$H_{\mathrm{TC}}|gg,1\rangle = \hbar\omega_c|gg,1\rangle + \sqrt{2}\hbar g\,|S,0\rangle$$

$$H_{\mathrm{bright}} = \hbar\begin{pmatrix}\omega_a & \sqrt{2}g\\ \sqrt{2}g & \omega_c\end{pmatrix}$$

On resonance, $\omega_a = \omega_c$, giving

$$H_{\mathrm{bright}} = \hbar\begin{pmatrix}\omega_c & \sqrt{2}g\\ \sqrt{2}g & \omega_c\end{pmatrix}$$

This has the same form as the Jaynes–Cummings Hamiltonian, but with

$$g_{\mathrm{eff}} = \sqrt{2}g$$

collective enhancement

The corresponding collective Rabi angular frequency is

$$\Omega_S = 2\sqrt{2}g$$

The antisymmetric state $|A,0\rangle$ remains uncoupled.

Collectively Enhanced Rabi Oscillations

Suppose the atoms begin in the symmetric state $|S\rangle$, with no cavity photon:

$$|\psi(0)\rangle = |S, 0\rangle$$

On resonance, and ignoring a common overall phase,

$$|\psi(t)\rangle = \cos\left(\sqrt{2}gt\right)|S, 0\rangle - i\sin\left(\sqrt{2}gt\right)|gg, 1\rangle$$

The corresponding probabilities are

$$P_{S,0}(t) = \cos^2\left(\sqrt{2}gt\right), \qquad P_{gg,1}(t) = \sin^2\left(\sqrt{2}gt\right)$$

$$\boxed{\Omega_S = 2\sqrt{2}g}$$

From prev. lecture (Slide: Vacuum Rabi oscillations)

$$|\psi(t)\rangle = \cos(gt)|e, 0\rangle - i\sin(gt)|g, 1\rangle.$$

By comparison, one atom has $\Omega_1 = 2g$.

The $\sqrt{N}$ Collective Enhancement

Define the state with all atoms in their ground states:

$$|G\rangle = |g_1 g_2 \cdots g_N\rangle$$

The symmetric state containing one atomic excitation is

$$|S_N\rangle = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} |g_1 \cdots e_i \cdots g_N\rangle$$

The collective lowering operator gives

$$J_- |S_N\rangle = \sqrt{N}\, |G\rangle$$

Therefore,

$$H_I |S_N, 0\rangle = \sqrt{N}\hbar g\, |G, 1\rangle$$

$$\boxed{g_{\text{eff}} = \sqrt{N} g, \qquad \Omega_{\text{coll}} = 2\sqrt{N} g}$$

Symmetric states

Symmetric states are unchanged when any two atoms are exchanged: the excitations are shared equally and coherently across the ensemble.

$$|S_2\rangle = \frac{|eg\rangle + |ge\rangle}{\sqrt{2}}$$

$$|S_3\rangle = \frac{|egg\rangle + |geg\rangle + |gge\rangle}{\sqrt{3}}$$

$$|S_4\rangle = \frac{|eggg\rangle + |gegg\rangle + |ggeg\rangle + |ggge\rangle}{\sqrt{4}}$$

and so on…

Dicke States: A Collective Spin

Each two-level atom behaves mathematically like a spin-$\frac{1}{2}$ system.

The ensemble can therefore be described using collective angular-momentum states

For the fully symmetric states of N atoms,

$$|j,m\rangle, \qquad j=\frac{N}{2}, \qquad m=-j,-j+1,\ldots,j$$

The number of excited atoms is

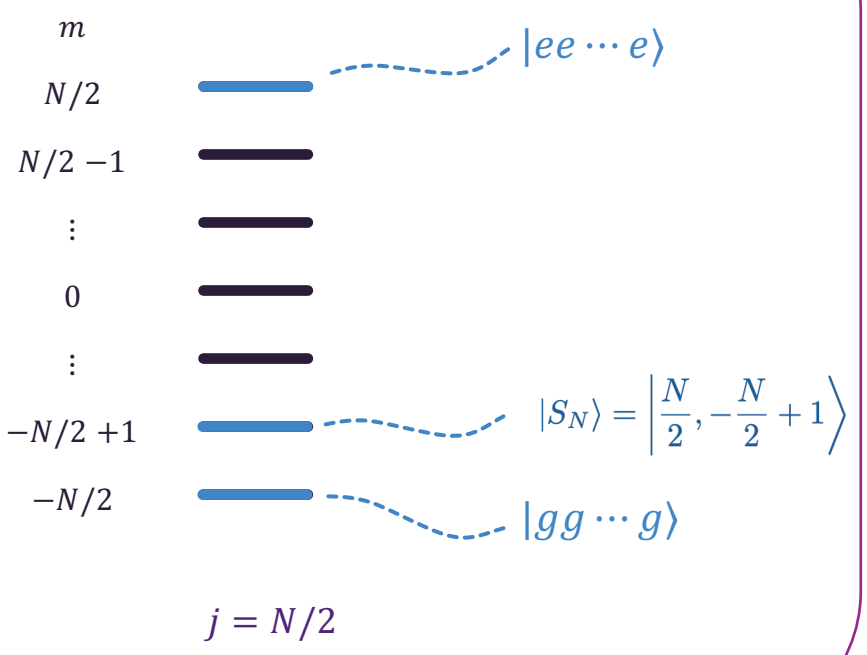
$$N_e = j+m$$

The states introduced previously can be written as

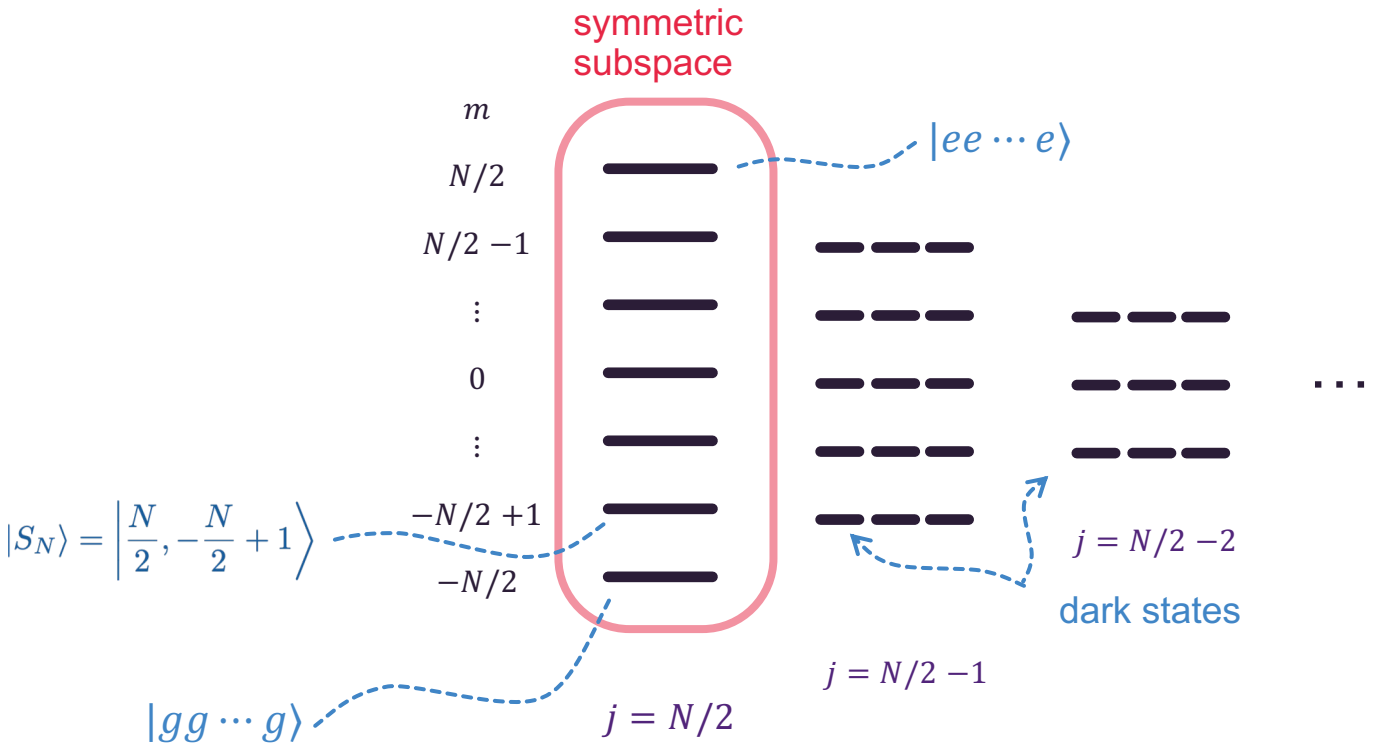
$$|G\rangle = \left|\frac{N}{2},-\frac{N}{2}\right\rangle = |gg\cdots g\rangle$$

$$|S_N\rangle = \left|\frac{N}{2},-\frac{N}{2}+1\right\rangle = \frac{1}{\sqrt{N}}\left(|eg\cdots g\rangle + |ge\cdots g\rangle + \cdots + |gg\cdots e\rangle\right)$$

Fully symmetric Dicke ladder: j = $N/2$



Beyond the Fully Symmetric Manifold



The full N-atom space contains several total-spin sectors:

$$j = \frac{N}{2}, \frac{N}{2} - 1, \ldots, j_{\min}$$

$j = N/2$ contains the fully symmetric Dicke states.

Smaller j sectors contain states that are not fully symmetric and generally show weaker collective emission.

$J_\pm$ moves the system vertically within a ladder but does not change j.

The lowest states of the smaller-j ladders can be dark:

$$J_- |j, -j\rangle = 0$$

For identical coupling and a symmetric initial state, the system remains in the $j = N/2$ manifold.

Moving Along the Dicke Ladder

The collective operators add or remove one atomic excitation:

$$\hat{J}_-|j,m\rangle = \sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle$$

$$\hat{J}_+|j,m\rangle = \sqrt{(j-m)(j+m+1)}\,|j,m+1\rangle$$

$\hat{J}^-$ removes one atomic excitation.

$\hat{J}_+$adds one atomic excitation.

These operators change m, but leave j unchanged.

For the symmetric single-excitation state,

$$\hat{J}_-|S_N\rangle = \sqrt{N}\,|G\rangle$$

The N indistinguishable atomic amplitudes add coherently, producing the collective enhancement $\sqrt{N}$.

From Tavis–Cummings to Superradiant Decay

Tavis–Cummings is a model of atom–field coupling.
Superradiance is a physical phenomenon of collectively enhanced emission.

Tavis–Cummings model
Question: How does an atomic ensemble exchange excitation with a selected field mode?

Dicke superradiant-decay regime
Question: How do the atomic excitation and emitted intensity evolve in time?

Both are governed by

$$\hat{J}_- = \sum_{i=1}^{N} \hat{\sigma}_-^{(i)}$$

What Is Superradiance?

Superradiance is collectively enhanced emission from a coherent atomic ensemble.

The emission is governed by

$$\hat{J}_- = \sum_{i=1}^{N} \hat{\sigma}_-^{(i)}$$

- The photon does not identify which atom emitted it.
- The atomic emission amplitudes can therefore add coherently.

Emission Rate of a Dicke State

Using the collective ladder matrix element,

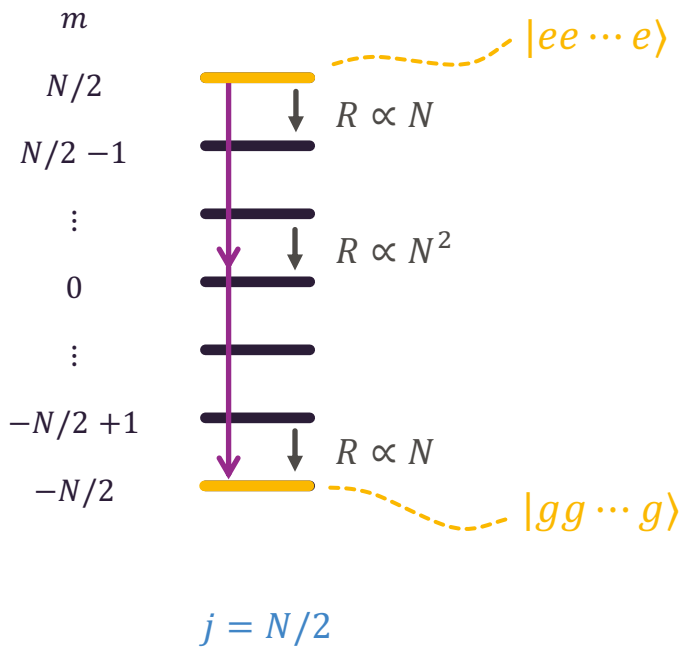
$$R_{j,m} = \Gamma \left| \langle j, m-1 | \hat{J}_- | j, m \rangle \right|^2 = \Gamma(j+m)(j-m+1)$$

For the fully symmetric manifold, $j = N/2$and $N_e = N/2 + m$, giving

$$\boxed{R = \Gamma N_e (N - N_e + 1)}$$

- Γ: single-atom spontaneous-emission rate
- N_e: number of excited atoms
- The additional factor $N - N_e + 1$ is the collective enhancement

The Superradiant Burst



The ensemble cascades down the symmetric Dicke ladder:

$$|j, j\rangle \longrightarrow |j, j-1\rangle \longrightarrow \cdots \longrightarrow |j, -j\rangle$$

Initially,

$$R_{\text{initial}} = N\Gamma$$

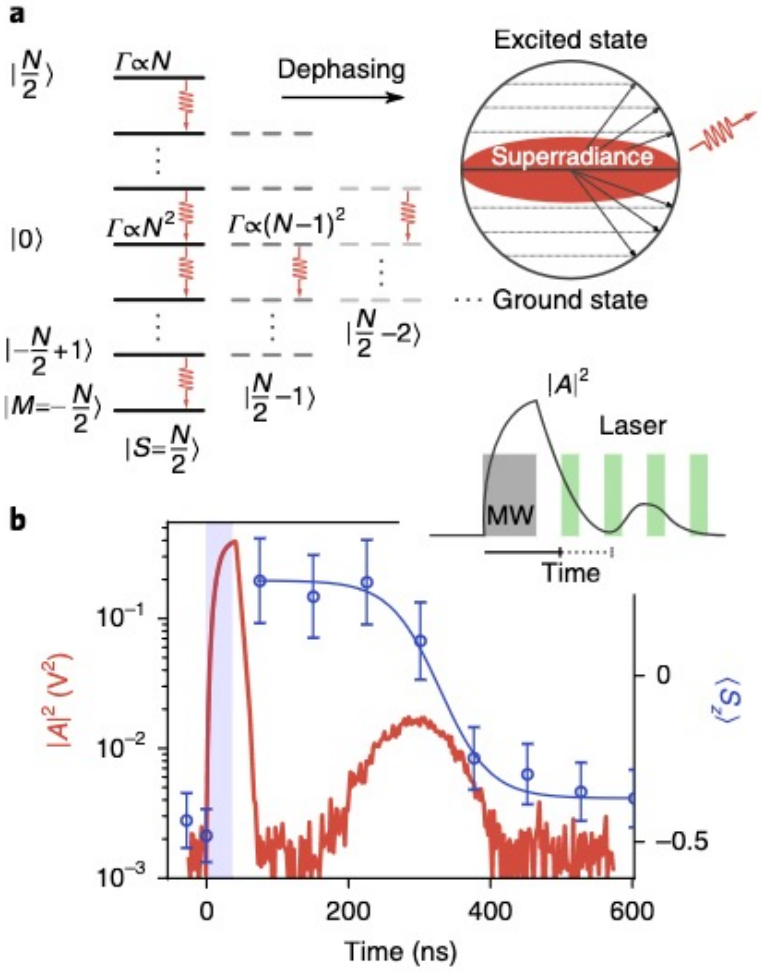
Near half excitation,

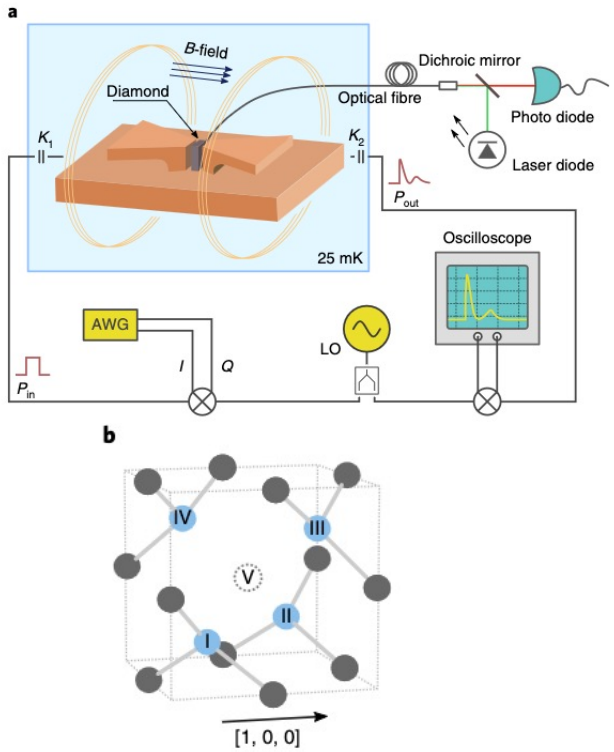
$$R_{\text{peak}} \approx \frac{\Gamma N^2}{4}$$

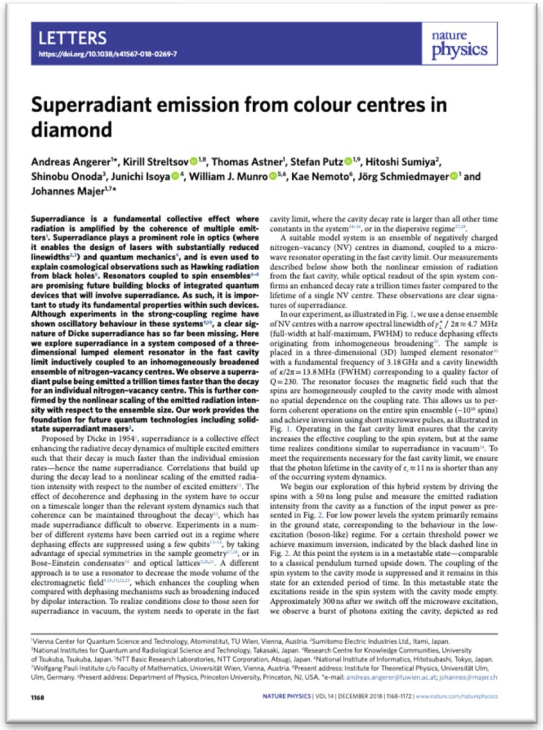
The emission rate first increases and then decreases.
This produces a short, intense superradiant pulse.
The N^2scaling occurs near the peak, not at the beginning.

Requires: coherent coupling to common field modes.
Dephasing or distinguishable emission suppresses the burst.

A large-N demonstration of superradiance in a solid-state spin ensemble







LETTERS

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Superradiant emission from colour centres in diamond

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Superradiance is a fundamental collective effect where radiation is amplified by the coherence of multiple emitters[1]. Superradiance plays a prominent role in optics (where it enables the design of lasers with substantially reduced linewidths[2–5]) and quantum mechanics[1], and is even used to explain cosmological observations such as Hawking radiation from black holes[6]. Resonators coupled to spin ensembles[7–11] are promising future building blocks of integrated quantum devices that will involve superradiance. As such, it is important to study its fundamental properties within such devices. Although experiments in the strong-coupling regime have shown oscillatory behaviour in these systems[12,13], a clear signature of Dicke superradiance has so far been missing. Here we explore superradiance in a system composed of a three-dimensional lumped element resonator in the fast cavity limit inductively coupled to an inhomogeneously broadened ensemble of nitrogen-vacancy centres. We observe a superradiant pulse being emitted a trillion times faster than the decay for an individual nitrogen-vacancy centre. This is further confirmed by the nonlinear scaling of the emitted radiation intensity with respect to the ensemble size. Our work provides the foundation for future quantum technologies including solid-state superradiant masers[3].

Proposed by Dicke in 1954[1], superradiance is a collective effect enhancing the radiative decay dynamics of multiple excited emitters such that their decay is much faster than the individual emission rates—hence the name superradiance. Correlations that build up during the decay lead to a nonlinear scaling of the emitted radiation intensity with respect to the number of excited emitters[1]. The effect of decoherence and dephasing in the system have to occur on a timescale longer than the relevant system dynamics such that coherence can be maintained throughout the decay[1], which has made superradiance difficult to observe. Experiments in a number of different systems have been carried out in a regime where dephasing effects are suppressed using a few qubits[14–17], by taking advantage of special symmetries in the sample geometry[12,18], or in Bose–Einstein condensates[19] and optical lattices[20,21]. A different approach is to use a resonator to decrease the mode volume of the electromagnetic field[4,10,11,22,23], which enhances the coupling when compared with dephasing mechanisms such as broadening induced by dipolar interaction. To realize conditions close to those seen for superradiance in vacuum, the system needs to operate in the fast cavity limit, where the cavity decay rate is larger than all other time constants in the system[24–26], or in the dispersive regime[27,28].

A suitable model system is an ensemble of negatively charged nitrogen–vacancy (NV) centres in diamond, coupled to a microwave resonator operating in the fast cavity limit. Our measurements described below show both the nonlinear emission of radiation from the fast cavity, while optical readout of the spin system confirms an enhanced decay rate a trillion times faster compared to the lifetime of a single NV centre. These observations are clear signatures of superradiance.

In our experiment, as illustrated in Fig. 1, we use a dense ensemble of NV centres with a narrow spectral linewidth of $\gamma_2^*/2\pi \approx 4.7$ MHz (full-width at half-maximum, FWHM) to reduce dephasing effects originating from inhomogeneous broadening[29]. The sample is placed in a three-dimensional (3D) lumped element resonator[30] with a fundamental frequency of 3.18 GHz and a cavity linewidth of $\kappa/2\pi = 13.8$ MHz (FWHM) corresponding to a quality factor of $Q = 230$. The resonator focuses the magnetic field such that the spins are homogeneously coupled to the cavity mode with almost no spatial dependence on the coupling rate. This allows us to perform coherent operations on the entire spin ensemble ($\sim 10^{10}$ spins) and achieve inversion using short microwave pulses, as illustrated in Fig. 1. Operating in the fast cavity limit ensures that the cavity increases the effective coupling to the spin system, but at the same time realizes conditions similar to superradiance in vacuum[31]. To meet the requirements necessary for the fast cavity limit, we ensure that the photon lifetime in the cavity of $\tau_c \approx 11$ ns is shorter than any of the occurring system dynamics.

We begin our exploration of this hybrid system by driving the spins with a 50 ns long pulse and measure the emitted radiation intensity from the cavity as a function of the input power as presented in Fig. 2. For low power levels the system primarily remains in the ground state, corresponding to the behaviour in the low-excitation (boson-like) regime. For a certain threshold power we achieve maximum inversion, indicated by the black dashed line in Fig. 2. At this point the system is in a metastable state—comparable to a classical pendulum turned upside down. The coupling of the spin system to the cavity mode is suppressed and it remains in this state for an extended period of time. In this metastable state the excitations reside in the spin system with the cavity mode empty. Approximately 300 ns after we switch off the microwave excitation, we observe a burst of photons exiting the cavity, depicted as red

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Summary

$$\boxed{\text{Jaynes–Cummings} \longrightarrow \text{Tavis–Cummings} \longrightarrow \text{superradiance}}$$

- The Tavis–Cummings model extends atom–field coupling from one atom to an ensemble.
- Symmetric states couple collectively, producing enhancements such as $g\sqrt{N}$, while antisymmetric states can be dark.
- The same collective amplitudes can produce superradiant emission with a peak rate proportional to N^2.